

Improved Bounds for the Critical Parameter of the Multiplicative Hilbert Family

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Status and source problem

This is a `candidate_partial_likely_valid` result for Karl-Mikael Perfekt and Alexander Pushnitski, *On the spectrum of the multiplicative Hilbert matrix*, arXiv:1705.01959 [1].

For $a > 0$, the source considers the positive Helson matrix on $\ell^2(\mathbb{N})$

$$M(g_a)_{m,n} = \frac{1}{\sqrt{mn}(a + \log(mn))}, \quad m, n \geq 1.$$

Its Theorem 1.2 produces a critical number $a_* > 0$: if $a \geq a_*$, $M(g_a)$ has no eigenvalue above π , whereas if $0 < a < a_*$ it has exactly one simple eigenvalue above π . On PDF page 3 the authors say that they cannot explicitly compute a_* and give only

$$\frac{1}{\pi} \leq a_* \leq 2.$$

The same passage asks for an explicit diagonalization of $M(g_a)$ and of the multiplicative Hilbert matrix $M_2(g_0)$.

Main result

Theorem 1. *Let a_* be the critical parameter in [1, Theorem 1.2], and let γ denote the Euler–Mascheroni constant. Then*

$$\boxed{\frac{1}{3} < a_* \leq 2\gamma}.$$

Numerically, $2\gamma = 1.1544313298\dots$. Both inequalities strictly improve the source estimates.

Proof intuition

The two bounds use different representations. At the matrix level, a two-dimensional compression already detects an eigenvalue above π at $a = 1/3$; deliberately coarse rational bounds for π and $\log 2$ suffice to certify the sign of its characteristic polynomial.

For the upper bound, the source factors $M(g_a)$ through an integral Hankel operator with kernel

$$h_a(t) = e^{-at/2}\zeta(1+t).$$

The logarithmic derivative of $t\zeta(1+t)$ is a decreasing function whose limit at zero is γ . Thus $t\zeta(1+t) \leq e^{\gamma t}$. When $a \geq 2\gamma$, this cancels the exponential in h_a and leaves the Carleman majorant $1/t$, whose Hankel operator has norm π .

Upper bound

We use a recent real-axis zeta inequality.

Lemma 1 (Garcia–Grenié–Molteni [2]). *The function*

$$s \mapsto \frac{\zeta'(s)}{\zeta(s)} + \frac{1}{s-1}$$

is decreasing on $(1, \infty)$.

This is Theorem 5 of arXiv:2607.08342. Define

$$\Phi(t) = \log(t\zeta(1+t)), \quad t > 0.$$

The Laurent expansion $\zeta(1+t) = t^{-1} + \gamma + O(t)$ gives $\Phi(0+) = 0$ and $\Phi'(0+) = \gamma$. Moreover,

$$\Phi'(t) = \frac{1}{t} + \frac{\zeta'(1+t)}{\zeta(1+t)}.$$

Lemma 1 says that Φ' is decreasing. Therefore

$$\Phi(t) \leq \gamma t, \quad \text{equivalently} \quad t\zeta(1+t) \leq e^{\gamma t} \text{quad}(t > 0). \quad (1)$$

The source's Theorem 2.2 identifies the nonzero part of $M(g_a)$ unitarily with the integral Hankel operator $H(h_a)$ on $L^2(\mathbb{R}_+)$, where $h_a(t) = e^{-at/2}\zeta(1+t)$. If $a \geq 2\gamma$, then

$$0 < h_a(t) \leq e^{-\gamma t} \frac{e^{\gamma t}}{t} = \frac{1}{t}.$$

The source's Carleman estimate (Lemma 2.1) consequently gives $\|H(h_a)\| \leq \pi$, hence $\|M(g_a)\| \leq \pi$. There can be no eigenvalue above π , so the defining dichotomy for a_* yields

$$a_* \leq 2\gamma. \quad (2)$$

This use of pointwise Carleman domination cannot improve the constant:

$$e^{-at/2}t\zeta(1+t) = 1 + (\gamma - a/2)t + O(t^2),$$

so the needed pointwise inequality fails near zero for every $a < 2\gamma$.

Lower bound

Set $a = 1/3$ and compress $M(g_a)$ to $\text{span}\{e_1, e_2\}$. Its matrix is

$$A = \begin{pmatrix} 3 & \frac{1}{\sqrt{2}(1/3 + \log 2)} \\ \frac{1}{\sqrt{2}(1/3 + \log 2)} & \frac{1}{2(1/3 + \log 4)} \end{pmatrix}.$$

We show that $\det(\pi I - A) < 0$. Since $3 < \pi < 22/7$ and the lower-right entry of A is positive,

$$(\pi - 3) \left(\pi - \frac{1}{2(1/3 + \log 4)} \right) < \pi(\pi - 3) < \frac{22}{49}.$$

Also $\log 2 < 7/10$. For completeness, the latter follows from

$$e^{7/10} > 1 + \frac{7}{10} + \frac{(7/10)^2}{2} + \frac{(7/10)^3}{6} > 2.$$

Consequently,

$$A_{12}^2 = \frac{1}{2(1/3 + \log 2)^2} > \frac{1}{2(31/30)^2} = \frac{450}{961} > \frac{22}{49}.$$

It follows that

$$\det(\pi I - A) = (\pi - A_{11})(\pi - A_{22}) - A_{12}^2 < 0.$$

For a real symmetric 2×2 matrix this says that its two eigenvalues straddle π ; in particular $\lambda_{\max}(A) > \pi$. The maximal Rayleigh quotient of the full operator $M(g_{1/3})$ is therefore greater than π . By the spectral dichotomy in [1, Theorem 1.2], $M(g_{1/3})$ has its simple eigenvalue above π , and hence

$$\frac{1}{3} < a_*. \quad (3)$$

Combining (2) and (3) proves Theorem 1.

Verification, upgrade attempts, and limitations

The lower estimate is a finite exact certificate; no floating-point spectral calculation enters the proof. The only external input beyond the source operator theory is Lemma 1. Its variable substitution is exact: $s = 1 + t$ makes its left-hand side equal to $\Phi'(t)$.

Before locating Lemma 1, an elementary midpoint estimate for the convex function x^{-1-t} gave

$$t\zeta(1+t) \leq t + (3/2)^{-t} \leq e^{(1-\log(3/2))t},$$

and hence the weaker new upper bound $a_* \leq 2(1 - \log(3/2))$. The monotonicity theorem upgrades this to the method-sharp constant 2γ .

The packet does not determine a_* , compute the eigenvalue $\lambda(a)$, or explicitly diagonalize either matrix. Higher finite compressions can numerically strengthen the lower endpoint, but no similarly transparent closed certificate was retained here. The bounded-symbol endpoint mentioned on the same source page is a distinct question already covered by the run attempt 2005.11951_multiplicative_hilbert_h1_endpoint_attempt.md.

A bounded novelty search used the exact phrases `critical value multiplicative Hilbert matrix`, `M(g_a) critical value`, the source arXiv id, and the constants in the preliminary bound. It found the source estimate but no later improvement. The July 2026 zeta paper was found through a separate search for real-axis bounds on $(s-1)\zeta(s)$; no application of its Theorem 5 to this Helson family was located. Novelty confidence is moderate. Human review should focus on the use of the source's critical dichotomy at $a = 1/3$ and on the bibliographic novelty check.

References

- [1] K.-M. Perfekt and A. Pushnitski, *On the spectrum of the multiplicative Hilbert matrix*, Ark. Mat. 56 (2018), 163–183; arXiv:1705.01959.
- [2] S. R. Garcia, L. Grenié, and G. Molteni, *Explicit upper and lower bounds on certain functions related to $\zeta(s)$ for $s > 1$* , arXiv:2607.08342 (2026).